



Spatial transcriptomics data and analytical methods: An updated perspective

Danishuddin ^{1,*†}, Shawez Khan ^{2,†}, Jong Joo Kim ^{1,*}

¹ Department of Biotechnology, Yeungnam University, Gyeongsan, Gyeongbuk 38541, Korea

² National Center for Cancer Immune Therapy (CCIT-DK), Department of Oncology, Copenhagen University Hospital, Herlev, Denmark

Spatial transcriptomics (ST) is a newly emerging field that integrates high-resolution imaging and transcriptomic data to enable the high-throughput analysis of the spatial localization of transcripts in diverse biological systems. The rapid progress in this field necessitates the development of innovative computational methods to effectively tackle the distinct challenges posed by the analysis of ST data. These platforms, integrating AI techniques, offer a promising avenue for understanding disease mechanisms and expediting drug discovery. Despite significant advances in the development of ST data analysis techniques, there is an ongoing need to enhance these models for increased biological relevance. In this review, we briefly discuss the ST-related databases and current deep-learning-based models for spatial transcriptome data analyses and highlight their roles and future perspectives in biomedical applications.

Keywords: spatial transcriptomics; deep learning; spatial omics databases; disease modeling

Introduction

The success of next-generation sequencing (NGS) has revolutionized the omics field.^(p1) This has led to the advancement of technologies, including single-cell RNA sequencing (scRNA-seq), which reveals complicated and uncommon cell populations, uncovering the regulatory links between genes.^{(p1),(p2)} Over the years, scRNA-seq technology combined with other omics data has undergone exponential growth, providing the insights into the mechanisms of homeostatic physiology and complex pathologies.^(p3) This method has facilitated accurate gene expression mapping to distinct cell types. Despite advantages, scRNA-seq has limitations, notably its inability to capture crucial spatial information, such as the precise location of cells. These limitations prompted the development of spatial transcriptomics (ST) technologies, which provides a thorough understanding of how cells are arranged in space and how they express their genes, thereby elucidating the complex molecular architecture of

biological systems.^(p4) Spatial information regarding cell location is important for understanding cell-cell communication, which is of great significance to tissue development.^{(p5),(p6)} ST offers exceptional insights that surpass the limitations of non-spatial single-cell transcriptomic data, like scRNA-seq. This is because the processes through which cells develop into tissues and interact with one another heavily rely on their interactions with the surrounding environment.

The emergence of ST has contributed to the generation of extensive knowledge in various domains, including, cancer, neuroscience and other diseases.^{(p7),(p8),(p9)} This enables the identification of specific cell types within the spatial context and the molecular features that differentiate normal and tumor tissues.^(p7) ST technologies can be broadly categorized into two main methods: (i) imaging-based; and (ii) sequencing-based technologies.^{(p10),(p11)} Image-based ST methods like merFISH^(p12) and seqFISH+^(p13) frequently achieve single-cell or subcellular

* Corresponding authors. Danishuddin, (danish23@yu.ac.kr), Kim, J.J. (kimjj@ynu.ac.kr).

† Equal contribution.

level resolution. Whereas sequencing-based technologies mainly focus on capturing gene expression patterns through HTS methods. These approaches, exemplified by techniques like Slide-seq^(p14) and Visium,^(p10) provide a broader spatial context of gene expression within tissue samples; thus, offering valuable insights into the distribution of transcripts across different regions.

The combination of ST and scRNA-seq data is an innovative approach for gaining in-depth, dynamic insights into the molecular processes underlying diseases.^(p15) This approach is recommended for efficiently diagnosing diseases, tracking disease progression and therapeutic effects, and discovering new categories of spatially targeted drugs.^(p16) Spatial biology platforms have the potential to accelerate translational drug discovery by enabling comprehensive phenotypic immunoassays, characterizing tissue structures, investigating tumor heterogeneity and assessing biomarker expression in disease-affected regions. This allows precise profiling of disease states and heterogeneity. Zhang *et al.*^(p17) extensively analyzed hepatocellular carcinoma (HCC) in the tumor microenvironment (TME) using ST. Among 15 HCC patients studied, five responded positively to neoadjuvant cabozantinib and nivolumab drug treatment followed by surgical resection. In the responsive tumors, ST profiling revealed increased immune cell and cancer-associated fibroblast (CAF) populations, indicating heightened proinflammatory signaling. Their findings suggest a spatially defined distribution of the immune response within the TME, with particular emphasis on the involvement of B cells as potential initiators of tumor cell cytotoxicity and orchestrators of the recruitment of other effector immune cells.

The increasing focus on harnessing single-cell data has driven the development of pipelines aimed at deepening our insights into the TME in cancer research. For example, researchers have introduced a single-cell guided pipeline to aid repurposing of drugs (ASGARD).^(p18) ASGARD uses a drug scoring system that considers all cell clusters to address intercellular heterogeneity within each patient, making it a superior choice for predicting the efficacy of single-drug therapies compared with traditional bulk-cell-based methods.

ST databases

The increase in ST data and the development of databases for biomolecule interactions provide many essential resources for understanding of biological systems, uncovering complex diseases and exploring potential therapeutic drugs. In recent years, several ST databases have been developed (Table 1). For instance, SpatialDB^(p19) is a carefully curated database containing 24 datasets derived from five different species and produced using eight distinct ST methods. The Spatial Transcript Omics Analysis Resource (SOAR) database^(p20) is a comprehensive repository, comprising 1735 samples sourced from 145 distinct datasets (accessed 19 Oct 2023). It covers 23 tissue types from various organs, developmental phases and pathological states, including cancer. Additionally, it also provides user-friendly interactive web interfaces designed to visualize spatial gene expression patterns, investigate the spatial diversity of genes and evaluate interactions between cells using unique, internally developed methods. Another database: Spatial Exploration of Cellular

Environments and Relationships (SPASCER),^(p21) is a comprehensive resource aimed at unravelling the diversity in tissue structures, the unique microenvironments of various tissue regions and the interactions between cells within different tissue architectures at various scales. SPASCER includes data from 43 different research studies, encompassing a total of 1082 datasets derived from 16 organ types (19 Oct 2023). These datasets combine scRNA-seq data with ST to enable deconvolution and mapping of cellular locations. Furthermore, SPASCER facilitates the analysis of cell–cell interactions, gene expression patterns and pathway enrichment, providing valuable insights into tissue organization and function.

Spatial TranscriptOmics DataBase (STOmicsDB)^(p22) is a versatile and user-friendly database that comprises 218 meticulously curated spatial transcript datasets spanning 17 different species (19 Oct 2023). It offers a range of features, such as literature searching, visualization of publicly available datasets, customized databases, data archiving and web tools. Aquila^(p23) is a database that accesses and analyzes spatial omics data. Currently, it stores 107 datasets from 30 diseases, including > 6500 regions of interest and 15.7 million cells or spots (19 Oct 2023). In addition, it also offers visualization options for spatial omics data, including representations of spatial cell distribution, spatial expression patterns and the co-localization of specific markers. A comprehensive repository of spatial transcriptomics (CROST)^(p24) serves as a repository of ST data, housing 182 datasets from various species, organs and diseases. It includes 1033 sub-datasets and identifies 48 043 genes related to tumors and their spatial variability. It also offers a standardized data processing pipeline, incorporates single-cell RNA sequencing and deconvoluted ST data. CROST integrates multi-omics data to investigate the roles of spatially variable genes in cancer progression and prognosis, providing a comprehensive understanding of their significance. The museum^(p25) of ST serves as a valuable repository for study-level metadata associated with a variety of ST datasets. Although these databases are developed for visualizing, depositing and analyzing spatial data, there is currently no online platform specifically dedicated to the spatial mapping of ST data. To fill this gap, Li *et al.*^(p26) developed Stellaris – a web server come database that aims for fast and accurate spatial mapping of user-uploaded scRNA-seq data. Stellaris implements a retrieval system that can assign spatial coordinates to single cells within tissue architecture. Stellaris is built on 101 curated ST datasets, 823 sections of various organs, developmental stages and pathological states in humans and mice.

Deep learning and machine learning methods for ST analyses

The recent progress made in ST has provided an opportunity to gain insights into how tissues are structured and how they function within their spatial environment.^{(p11),(p27),(p28)} Data produced by ST technologies is inherently characterized by noise, high dimensionality, sparsity and multimodality, necessitating the use of specialized computational tools to ensure precise and resilient analysis. Advances in AI or other computational approaches (Figure 1) enable researchers to capitalize on the

TABLE 1

Databases for spatial omics-related data.

Database	Description	Year	Web address	Refs
SpatialDB	24 datasets derived from five different species and produced using eight distinct spatial transcriptomics methods	2020	https://www.spatialomics.org/SpatialDB	(p19)
SOAR	1735 samples sourced from 145 distinct datasets	2022	https://soar.fsm.northwestern.edu/	(p20)
	Integrated user-friendly interactive web interfaces to visualize spatial gene expression patterns, spatial diversity of genes and cell–cell interaction			
SPASCEr	Data collected from 43 different research studies, encompassing a total of 1082 datasets derived from 16 organ types	2022	https://ccsm.uth.edu/SPASCEr/	(p21)
	Also facilitates the analysis of cell–cell interactions, gene expression patterns and pathway enrichment			
STOmicsDB	218 meticulously curated spatial transcript datasets spanning 17 different species	2022	https://db.cngb.org/stomics/	(p22)
	Also provide facilities like: literature search, visualization of publicly available datasets, data archiving and online analysis tools			
Aquila	107 datasets from 30 diseases, including 6500 + regions of interest and 15.7 million cells or spots	2023	https://aquila.cheunglab.org	(p23)
CROST	Stored 1033 samples, which include expression and spatial coordinate information	2023	https://ngdc.cncb.ac.cn/crost/home	(p24)
	Implemented a standardized analysis pipeline that leverages raw sequencing data			
Museum	A comprehensive literature review of spatial transcriptomics	2022	https://github.com/pachterlab/LP_2021	(p25)
	This includes an examination of experimental techniques, studied species and tissues, as well as the computational approaches employed			
STellaris	A web server come database to publicly accessible spatial transcriptomics (ST) data	2023	https://spatial.rhesusbase.com	(p26)
	101 manually selected ST datasets with 823 sections spanning various organs, stages of development and disease conditions in mice and humans			
STASH	Facilitate researchers in promptly identifying the suitable tool for their requirements or even offer guidance to those interested in improving tool development	2023	https://cailab.labshare.cn:7004	(p34)

Information accessed on 19 October 2023.

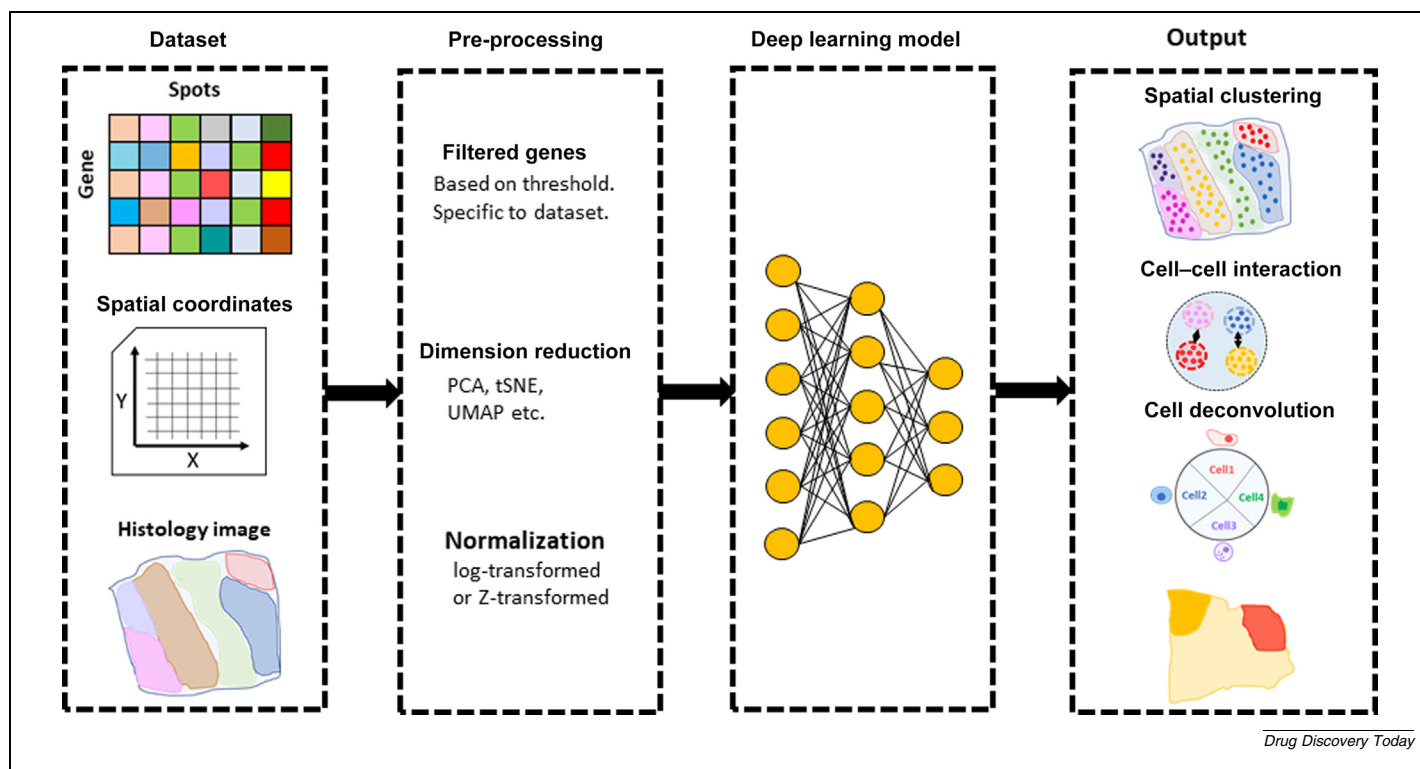
increasingly large and complex ST datasets to facilitate biological discovery.^{(p29),(p30)}

Deep-learning-based technologies have opened new doors in biological-based applications, unfolding complex phenomenon by enabling the analysis and interpretation of vast biological datasets with unprecedented accuracy and efficiency.^{(p31),(p32),(p33)} These integrative computational techniques that combine ST data with other data modalities have had a substantial impact (Figure 2). They have not only addressed the primary challenges encountered in current ST technologies but have also provided profound biological insights into the intricate and diverse organization of tissues. These methods are used more frequently in integrated experimental and computational approaches, where they are most useful in the separation of meaningful patterns from complex biological systems. Although great progress has been made in the development of various ST analytical methods (Table 2), the task of selecting and efficiently utilizing suitable analysis tools has become increasingly complex. Considering this limitation, Han *et al.* developed STASH^(p34) – a user-friendly database of ST analysis tools, aiming to simplify the process of selecting the appropriate tool for the task. They meticulously

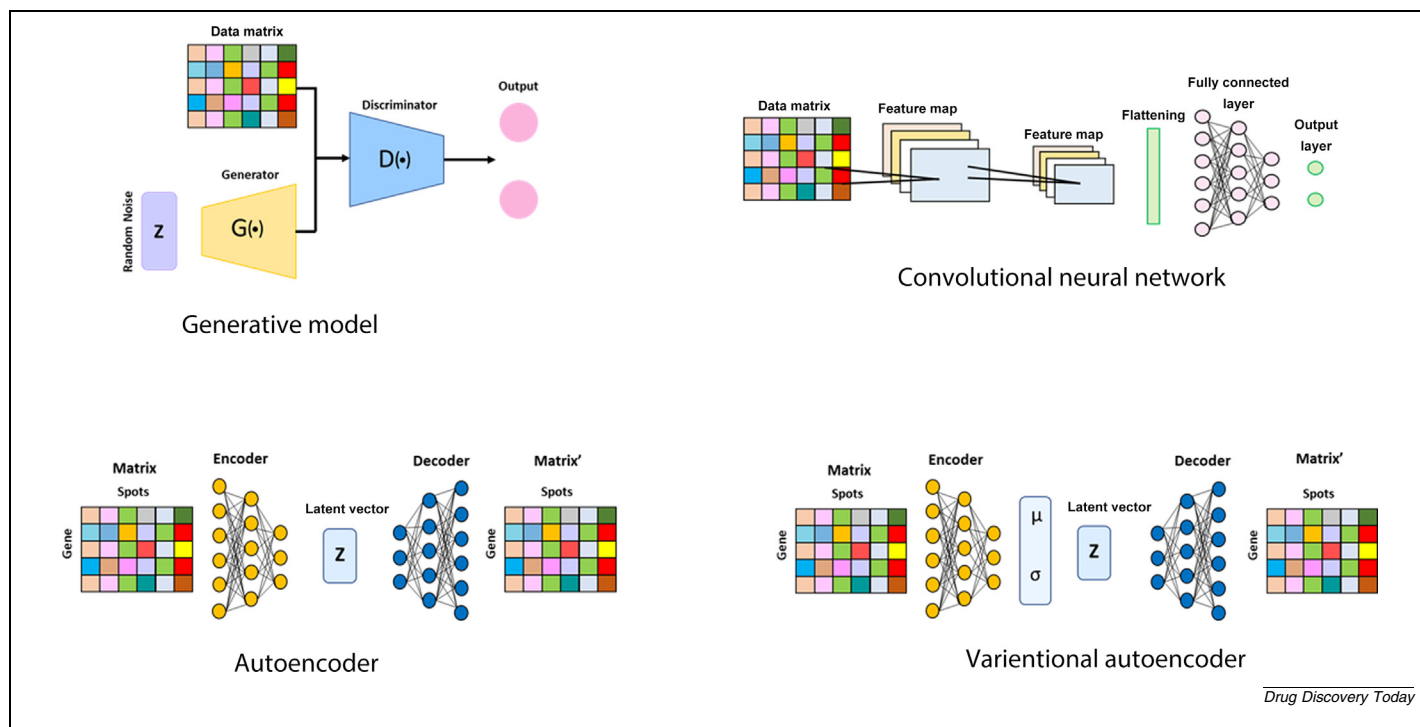
gathered and categorized the majority of currently available tools, offering valuable insights into their current status and emerging trends.

Spatial clustering

Clustering is one of the significant steps in ST data analyses.^(p35) This process enables the clustering of genes with similar transcriptional and spatial profiles. Clustering is a vital task for the annotation of cell types, understanding tissue structure composition, the identification of spatially variable gene identification and gene ontology (GO) analysis.^{(p35) (p36)} To date, several deep learning models have been developed that can incorporate expert domain knowledge about the statistical structure of these different regions in tissues. For example, Tan *et al.*^(p37) developed SpaCell, which uniquely combines gene expression data and tissue images to perform cell-type clustering. SpaCell is recognized as a pioneering approach to integrating gene expression data and tissue images within the context of cell-type clustering. StLearn^(p38) uses deep learning for image analysis and employs linear principal component analysis (PCA) for spatial gene expression feature extraction. SpaGCN^(p39) can effectively integrate gene expression, histology images and spatial coordinate

**FIGURE 1**

Deep learning framework in spatially resolved transcriptomics. This figure provides a visual summary of the various deep learning methods that can be used for spatial transcriptomics (ST) data analyses.

**FIGURE 2**

Overview of deep learning models and their application in spatial transcriptomics (ST) data analysis. The process involves the selection of raw data from available platforms. This is further followed by filter genes, dimension reduction and normalization, etc. The data obtained from this step can be adopted by any deep learning framework.

data using the graph convolutional network (GCN). This method aggregates gene expression data from neighboring spots, enabling the detection of spatial domains with consistent gene expression and histology.

ConGI^(p40) integrates spatial and expression information by aligning gene expression data with histopathological images using contrastive learning. This method employs three contrasting loss functions within each modality and between the two modalities, enabling the learning of shared representations. GraphST^(p41) uses a graph-based self-supervised contrasting learning model to extract meaningful and distinguishing representations for spots, revealing cell types in ST data. Although these spatial clustering techniques have led to significant progress in identifying spatial regions, they often assume that spots belonging to the same cell subpopulation are close in physical proximity.

Recently, Peng and colleagues introduced STGNNs^(p42) – an innovative spatial clustering technique used to identify spatial domains that combine a graph neural network (GNN), a denoising autoencoder and k-sums clustering. This method merges gene expression data and spatial context to create embedding features for individual spots using a deep graph convolutional network based on deep graph infomax. DeepST^(p43) uses a GNN autoencoder and a denoising autoencoder to construct the latent representation of augmented ST data to identify spatial domains. STMGCN^(p44) employs multi-view graph convolution networks (MGCN) to identify spatial domains in ST data. It generates distinct view-oriented embeddings by merging gene expression data with multiple neighbor graphs. STMGCN is proficient in identifying genes with spatially variable expressions featuring enriched patterns within the identified domains. DSSC^(p45) incorporates spatial information of individual cells into the clustering process using Kullback–Leibler (KL) divergence on the autoencoder's bottleneck layer.

Spatially variable genes

Identifying spatially variable genes (SVGs) is another important step in ST data analysis. SVGs have the potential to be a novel marker for distinct cell types, offering opportunities to enhance expression histology and provide deeper insights into the spatial architecture of the data. Several machine-learning-based approaches have been designed to identify SVGs (Table 2). These methods leverage machine learning techniques to enhance the statistical framework, achieving data compression and reducing computational load, whereas methods like Tangram^(p46) and CytoSpace^(p47) leverage ST information to infer cell locations in scRNA-seq data. Other methods like SpaOTsc^(p48) aim to infer cell–cell interactions from scRNA-seq data but can struggle with large datasets. NovoSpaRc^(p49), which also employs optimal transport algorithms, assumes that nearby cells have similar gene expression profiles and can recover cell locations without a spatial reference. Zhang *et al.*^(p50) introduced CeLery – a machine-learning method designed for recovering cell locations in scRNA-seq data using gene expression and spatial relationships from ST data. CeLery excels at recovering 2D locations and spatial domains while providing uncertainty estimates. Hao *et al.* introduced SOMDE^(p51), which is a robust and scalable method for

identifying SVGs. This method employs a combination of a self-organizing map (SOM) neural networks and Gaussian process to characterize spatial data. STAMarker^(p52) is a novel tool designed to address key limitations in existing ST analysis methods. Unlike traditional methods that independently test genes, STAMarker employs a three-stage ensemble framework, combining multi-layer perceptron (MLP) classifiers, graph-attention autoencoders and saliency map computation. STAMarker offers a unique advantage: the ability to pinpoint spatial domain-specific SVGs, facilitating deeper insights into specific regions, which is crucial in understanding complex biological phenomena.

Detecting SVGs is similar to finding differentially expressed genes. It helps us to identify the spatial distribution of genes and related molecular processes within cellular environments. However, SVG expression alone does not capture all the spatial cell distribution details. To overcome this limitation, Meng-Ling *et al.*^(p53) introduced a novel approach called Spatially Informed Artificial Intelligence (SPIN-AI) that can identify spatially preferred genes (SPGs) without making any prior spatial assumptions. SPIN-AI learns to reconstruct ST profiles from gene expression data alone, highlighting potential SPGs involved in spatial regulatory mechanisms governing cellular distribution in the pathological microenvironment.

Cell-type deconvolution

A key question in the analysis of ST data is the understanding of cell-type diversity. Cell-type deconvolution is a crucial consideration in the processing of ST data that enables a better understanding of the heterogeneity, development and interaction of tissues and complex biological systems.^(p54) Machine learning plays a significant part for systematically understanding the effects of spatial variations on cellular identity, including mapping spatial distribution of cell types by integrating tissue-specific ST data.^(p11) These methods utilize supervised or unsupervised operations, either in combination with high-resolution H&E images or by integrating scRNA-seq data. For example, SPOTlight^(p55) utilizes a deconvolution algorithm based on non-negative matrix factorization regression and nonnegative least squares. It integrates reference scRNA-seq data to establish cell-type-specific topic profiles, which are then used to deconvolute spatial spots. This approach identifies cell states and subpopulations, elucidating ST data complexity. One of the major limitations of this method is that it does not effectively capture and utilize the intrinsic topological information regarding cell type compositions within spots, which is crucial for understanding the relationships between observed gene expression patterns and associated cell types in spots. Over the past few years, GCN have shown potential in leveraging the inherent topology of data to improve the performance of models. For instance, DSTG^(p56) is a semi-supervised spatial data deconvolution approach that leverages a GCN model. It begins by using scRNA-seq to generate pseudo ST data and then constructs a link graph that encodes the similarity between all spots in real and pseudo ST data. This framework utilizes a multilayer GCN, which takes input from the link graph and concatenated pseudo ST and real ST data, enabling the interpretation of gene expression embeddings and local graph structures.

TABLE 2

A summary of methods for spatially resolved transcriptomics data analysis.

Category	Method	Algorithms	Description	Refs
Spatial clustering and spatially variable genes	SpaGCN	Graph Convolutional Network	Use gene expression, spatial locations and histology data (optional) for clustering analysis	(p39)
	ConGI	Contrastive learning approach	Identify spatial domains and spatially variable genes Infer spatial domains by integrating gene expression and histopathological images	(p40)
	GraphST	Graph-based self-supervised contrastive learning	Perform multiple tasks including trajectory inference, clustering and visualization Identify cell types in spatial transcriptomics data by extracting significant and distinctive representations for spots	(p41)
	STGNNks	Graph Neural Network, denoising autoencoder and k-sums clustering	Identify cell types in spatial transcriptomics data	(p42)
	DeepST	Graph Neural Network	Identify spatial domain in spatial transcriptomics data	(p43)
	STMGCN	Graph Convolutional Networks	Identify spatial domains of spatially resolved transcriptomics	(p44)
	DSSC	Autoencoder and Kullback–Leibler (KL) divergence	Generate distinct view-oriented embeddings by merging gene expression data with multiple neighbor graph Perform clustering of scRNA-seq	(p45)
	SpaOTsc	Transport model system	Incorporates spatial information of individual cells into the clustering process Mapping between scRNA-seq data and spatial data	(p48)
			Infer cell–cell interactions from scRNA-seq data	
			Perform cell spatial subclustering	
	NovoSpaRc	Statistical transport model	Probabilistically assigns cells to tissue locations	(p49)
	CeLery	Supervised deep learning	Predicts the spatial locations of individual cells using only information derived from single-cell RNA sequencing data	(p50)
	SOMDE	Self-organizing map neural network and Gaussian algorithm	Recover cell locations in single-cell RNA-seq Identify spatially variable genes	(p51)
	STAMarker	Graph-attention autoencoders, multilayer perceptron	Identify spatial-domain-specific SVGs	(p52)
Cell-type deconvolution	SPIN-AI	Feedforward neural network	Identify spatially predictive genes (SPGs)	(p53)
	SPOTlight	Non-negative matrix factorization (NMF) regression	Predict the spatial distribution of cell types and states within a complex tissue	(p55)
	DSTG	Graph Convolutional Networks	Deconvolute the observed gene expressions at each spot of ST data	(p56)
			Shows excellent precision and consistent performance across various experimental protocols	
	SONAR	Poisson–Gamma mixture distribution	Cell-type deconvolution	(p57)
	EnDecon	Ensemble learning	Can accurately map region-specific cell types in real spatial transcriptomic data Enables cell type deconvolution of spatially resolved transcriptomics data Identifies multiple cell types on spots Provides valuable insights into spatial heterogeneity and regionalization of tissues	(p58)
	STdGCN	Graph Neural Network	Developed for cell type deconvolution	(p59)
			Integrate the expression profiles of single-cell data along with spatial localization information from ST data	

TABLE 2 (CONTINUED)

Category	Method	Algorithms	Description	Refs
Spatial interaction analysis	NCEM	Graph Neural Network	Modeling intercellular communication in tissues	(p62)
	GCNG	Graph Convolutional Neural Network	Supervised computational framework for inferring cell–cell or gene–gene interaction from spatial single-cell data	(p63)
	CLARIFY	Multi-level graph autoencoder (GAE)	Intracellular and extracellular interaction networks by leveraging the spatial organization of single cells provided by spatial transcriptomics data	(p64)
Enhancing gene expression	DEEPsc	Deep Neural Network	Impute spatial information onto a scRNA-seq dataset	(p66)
	DeepSpaCE	Deep learning method	Anticipating gene expression levels and identifying transcriptomic cluster types using tissue spot images	(p67)
	XFuse	Deep generative model	Combine <i>in situ</i> RNA capturing data with histology image data	(p68)
			Generate high-resolution transcriptomics images using generative models	
	DIST	Deep learning method	Perform various tasks, including imputation, clustering, differential expression analysis and functional enrichment analysis	(p69)
			Enhances the gene expression for original spots and imputed spots using self-supervised learning and transfer learning	

SONAR, developed by Zhiyuan *et al.*,^(p57) is a novel model designed for cell-type deconvolution with ST data. This method directly works with the raw count data from ST and employs a geographically weighted regression framework. EnDecon,^(p58) utilizing a weighted optimization model, improves accuracy by merging outcomes from various base deconvolution methods. STdGCN^(p59) is a graph neural network that incorporates the spatial localization information from the ST data and the expression profiles from single-cell data for cell-type deconvolution. Other methods such as Vesalius^(p60) employed image processing technology to unravel tissue anatomy from ST data. Uniquely, Vesalius identified regions consisting of diverse cell types and effectively reconstructed tissue structures in high-resolution ST data, encompassing the mouse brain, embryo, liver and colon.

Spatial interaction analysis

Understanding cell–cell communication is essential to unravel how cells engage and how such interconnected networks evolve in the context of specific diseases.^(p61) Deciphering the overall cell–cell communication landscape holds promise for devising targeted strategies. Many deep learning methods in this area have been developed. For example, node-centric expression models (NCEM)^(p62) is a deep generative framework designed to simulate cellular communication in tissue microenvironments. It creates a spatial graph based on Euclidean distances, incorporating gene expression data, cell type information and batch assignments as inputs. In other work, a graph convolutional neural network for genes (GCNG) has been proposed for deducing gene interactions in cell–cell communication within single-cell ST data.^(p63) GCNG takes two inputs: a gene expression matrix representing a gene pair and a matrix encoding the spatial graph from ST data. The framework of GCNG consists of two graph convolutional layers: a flattening layer, a dense layer and a final classification layer responsible for determining gene pair interactions. Bafna *et al.*^(p64) presented a novel technique called CLARIFY, which is a multilevel graph autoencoder (GAE). This method improves the representation of intracellular and extra-

cellular interaction networks by leveraging ST data. It takes ST data as input and generates embeddings at the cell and gene levels, capturing single-cell gene expression, spatial context and gene regulatory information. The model is designed to enhance the precision of reconstructing cell-level adjacencies while penalizing significant alterations in intracellular gene interactions.

Enhancing gene expression

Another crucial component of computational ST analysis is improving spatial gene expression.^(p65) These tasks often require high-resolution histological images or scRNA-seq data. Several models utilizing deep learning approaches have been developed to improve the resolution of ST data. For instance, DEEPsc^(p66) utilizes deep learning to transfer spatial information from an ST assay to scRNA-seq data from the same tissue. It calculates a probability score ranging from 0 to 1 for each scRNA-seq cell and spatial ST spot, indicating the likelihood of a cell being associated with a specific spot. To achieve this, DEEPsc employs a fully connected neural network that takes input from two vectors of equal length: one containing dimensionally reduced gene expression values for a cell and the other containing ‘features’ from each spatial spot. The model predicts the spatial location of a cell based on the matching likelihood between its single-cell and spatial feature vectors.

DeepSpaCE^(p67) is a semi-supervised learning approach designed to estimate spatial gene expression from histological images and enhance the resolution of ST data using a modified VGG16-based convolutional neural network. H&E images are divided into sub-images representing spatial spots, which are then processed by the neural network consisting of 16 weight layers. Bergenstrahle *et al.*^(p68) developed XFuse to infer high-resolution spatial gene expression from histology image data by integrating low-resolution gene expression from *in situ* sequencing with high-resolution histology images. XFuse operates under the assumption that the conditional distribution of gene expression data follows a negative binomial (NB) distribution, whereas histology images follow a Gaussian distribution. Zhao developed

a deep-learning-based approach called DIST,^(p69) which effectively imputes gene expression profiles for unmeasured locations and enhances gene expression in the originally measured spots and the imputed ones. This is achieved through a combination of self-supervised learning and transfer learning. The performance of DIST was assessed across various tasks, including imputation, clustering, differential expression analysis and functional enrichment analysis.

Challenges and future perspectives

The rapid growth of ST technology offers an unprecedented view of gene expression within the spatial context of tissue ecosystems. However, this growth comes with the share of challenges that must be addressed to fully harness its potential for advancing our knowledge of biology and disease. One of the primary challenges in ST is the complexity of tissues.^(p70) Cells in a tissue are highly specialized and their intricate relationships, spatial arrangements and functional diversity contribute to the overall complexity.^(p71) For example, different cell types can exhibit distinct gene expression signatures, and their interactions can vary spatially within the tissue. Accurately classifying these cell types is crucial for understanding tissue functionality, because different cell populations often contribute distinct roles to overall tissue function. Therefore, addressing this challenge is fundamental to unlocking the full potential of ST in advancing our understanding of tissue biology and its implications in health and disease.

Another significant challenge is the inherent spatial heterogeneity of tissues.^(p72) Understanding how gene expression varies across different regions of a tissue is essential for unravelling complex biological processes; but it demands sophisticated methods that can account for this heterogeneity. Integration of ST data with other omics or image data also poses another significant challenge. This integration is vital and holds the potential to provide a holistic view of tissue function and regulation. Several techniques have been developed for data integration; however, the majority of these focuses on single-cell RNA-seq datasets, thereby neglecting spatial information. Liu *et al.*^(p73) introduced PRECAST, a data integration technique designed for managing complex batch effects and biological variations in multiple ST datasets. It combines spatial factor analysis, spatial clustering and embedding alignment in a unified manner and is adaptable to ST datasets from various platforms. In analysis of ST data, the pre-processing phase plays a crucial part in shaping the outcomes. ST data are generated by sequencing the captured locations, resulting in a count matrix, often referred to as the gene expression matrix. When applying conventional analytical techniques, such as dimensionality reduction and statistical methods, it is important to use suitable transformation and normalization strategies.^(p74) Avoiding these steps can potentially result in misinterpretations of the data. Future research should consider the compositional nature of gene expression matrices and explore alternative transformation and normalization methods. These challenges necessitate the development of robust computational methods to address important biological questions concerning cell types, interactions and regional segmentation. Advancement in computational methods

has an important role in extracting meaningful biological signals from raw data. These methods, employed to analyze spatially resolved transcriptomic data, have the potential to identify and characterize the diversity of cells within their spatial environments. Deep learning methods have brought exceptional opportunities for enhancing our understanding the ST field, including spatial alignment, reconstruction, spatial clustering and the integration of ST datasets with other omics data.

Many of these ST analysis frameworks have been constructed based on these techniques, which have proven to be suboptimal in many applications owing to the scale, multiple modalities and constraints of spatially-resolved data.^(p75) Indeed, there are challenges associated with such technologies, including accuracy, interpretability, stability and computability.^(p76) Interpretability is an important step with which the results of a model or system can be understood and explained, facilitating meaningful insights and decision-making. In addition, stability enables the consistency and robustness of a model's performance across different datasets or under varying conditions, indicating reliability and dependability. Practical consideration of these concepts, along with their trade-offs in machine learning model choice, implementation and the development could enable a more effective integration of machine learning with scientific discovery. Additionally, owing to the noise and multi-modality of ST data, there exists an unmet need for methods that account for batch effects in spatial and gene expression data. Batch effects result from technical disparities between datasets, such as variations in scRNA-seq methods, platforms, technologies, data processing, quality control, normalization and even differences in donors, sample collection times and personnel handling the data. Although numerous methods have been created and proven effective at reducing the batch effects,^(p77) there is still substantial room for improvement. The innovations in biological methods will continue to spur the creativity in algorithm development, with an emphasis on deep-learning-based frameworks.

Concluding remarks

ST is a technology that enhances our understanding of complex biological systems and diseases, thus offering significant potential for advancement in these areas. However, the technology faces several challenges, including the complexity of tissues, spatial resolution limitations, integration, low quality data, data noise, sampling bias and technology diversity. Addressing these challenges through interdisciplinary collaborations, technological advancements and ethical frameworks is essential to unlock the full potential of ST and accelerate discoveries in biology and medicine. In this review, our goal was to provide readers with the recent development in deep learning approaches and explore ways to tackle the challenges presented by the ST domain. ST continues to analyze patient samples, finding novel therapeutic biomarkers and getting closer to personalized treatment for individuals. Although the availability of deep learning frameworks for ST analysis is currently limited, we anticipate that ongoing progress will remain pivotal in unlocking the potential of spatial technologies in analyzing, visualizing and understanding newly generated data. As these technologies continue to progress, they offer the potential to reveal more-profound

understanding of how cells behave in tissues, thereby enhancing the capabilities of biomedical sciences including biomarker discovery and drug development.

Author contributions

Danish and JJ: conception and design the work. Danish and SK: collected the information and prepared the first draft of the manuscript. Danish and JJ: further modified the manuscript. All authors made substantial contributions to analysis, editing and writing of the manuscript.

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